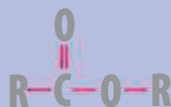
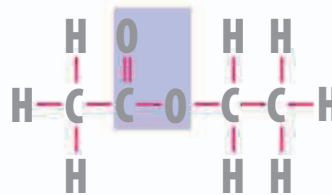


Esters

When a carboxylic acid reacts with an alcohol, they form an ester. The functional group of the ester links the two original molecules together. The fats and oils in living things—including the lipids that form cell membranes—are esters. Soaps, oils, and fats are also all types of ester.



△ **Functional group**
The oxygen from alcohol bonds to the carboxyl (carbon and oxygen group) from the acid.



△ **Ethyl ethanoate**

This ester has a strong pear drop smell and is used as nail varnish remover.



△ **Thiol smells**

The odor in garlic as well as the noxious smell sprayed by skunks is due to a thiol. Its functional group is called a sulfhydryl.



△ **Amine smells**

The smell of fish is due to the presence of a compound called trimethylamine. Its functional group includes nitrogen.

Thiols and amines

Thiols are similar to alcohols, except the functional group has a sulfur instead of an oxygen atom. The word “thiol” is a mixture of the Latin words for sulfur and alcohol. These compounds have strong smells. Amines are another smelly group of organic compounds. They have a functional group with one nitrogen and two hydrogen atoms. When an amine group attaches to a carboxylic acid, it forms an amino acid, one of the building blocks of proteins.

REAL WORLD

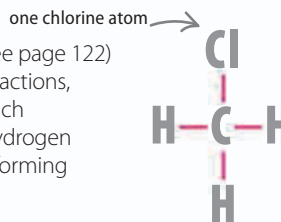
Ant stings

Some insect venoms, especially the stings of fire ants, have methanoic acid as their active ingredient. A fire ant squirts the acid onto attackers, causing small but painful burns. The original name, formic acid, is derived from the Latin word for “ant.”



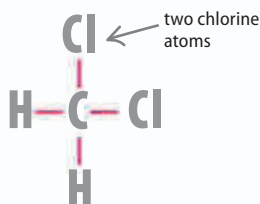
Organic halides

Members of the halogen group (see page 122) form only one bond in chemical reactions, just like hydrogen, but they are much more reactive. Halogens replace hydrogen atoms in hydrocarbon molecules, forming organic halides.



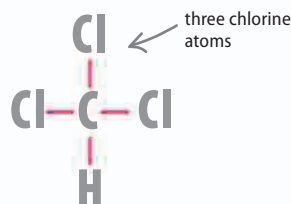
◁ **Chloromethane**

With just a single chlorine atom, this is the most reactive of this family of compounds. One of its uses is to make silicone rubbers.



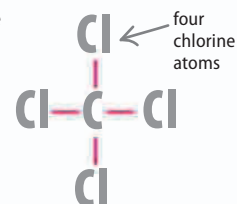
△ **Dichloromethane**

This sweet-smelling liquid is used in paint strippers, aerosol sprays, and to decaffeinate coffee.



△ **Trichloromethane**

Better known as chloroform, this compound was one of the first anesthetics.



△ **Tetrachloromethane**

Also known as carbon tetrachloride, this toxic liquid is banned in some countries.